Data structure

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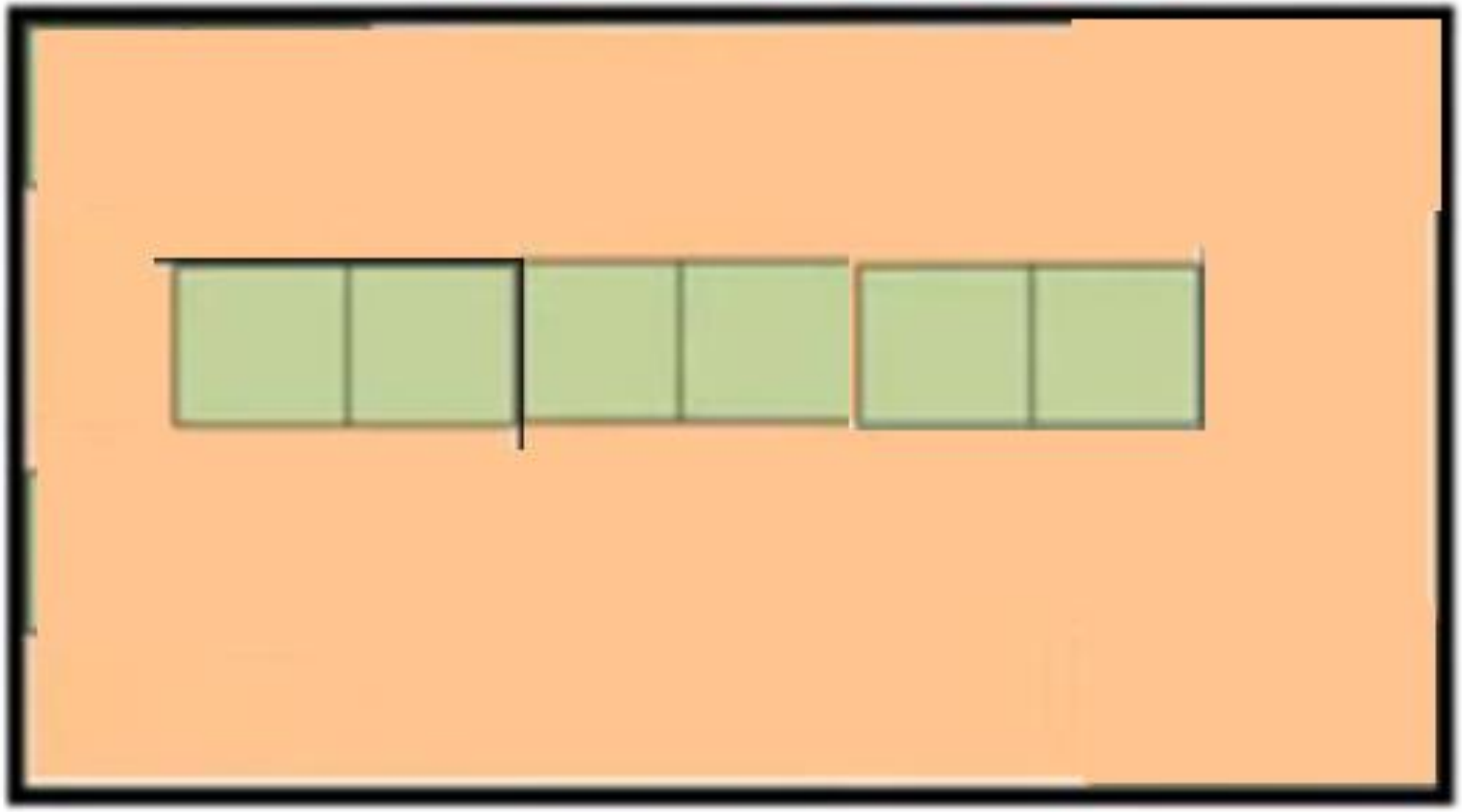
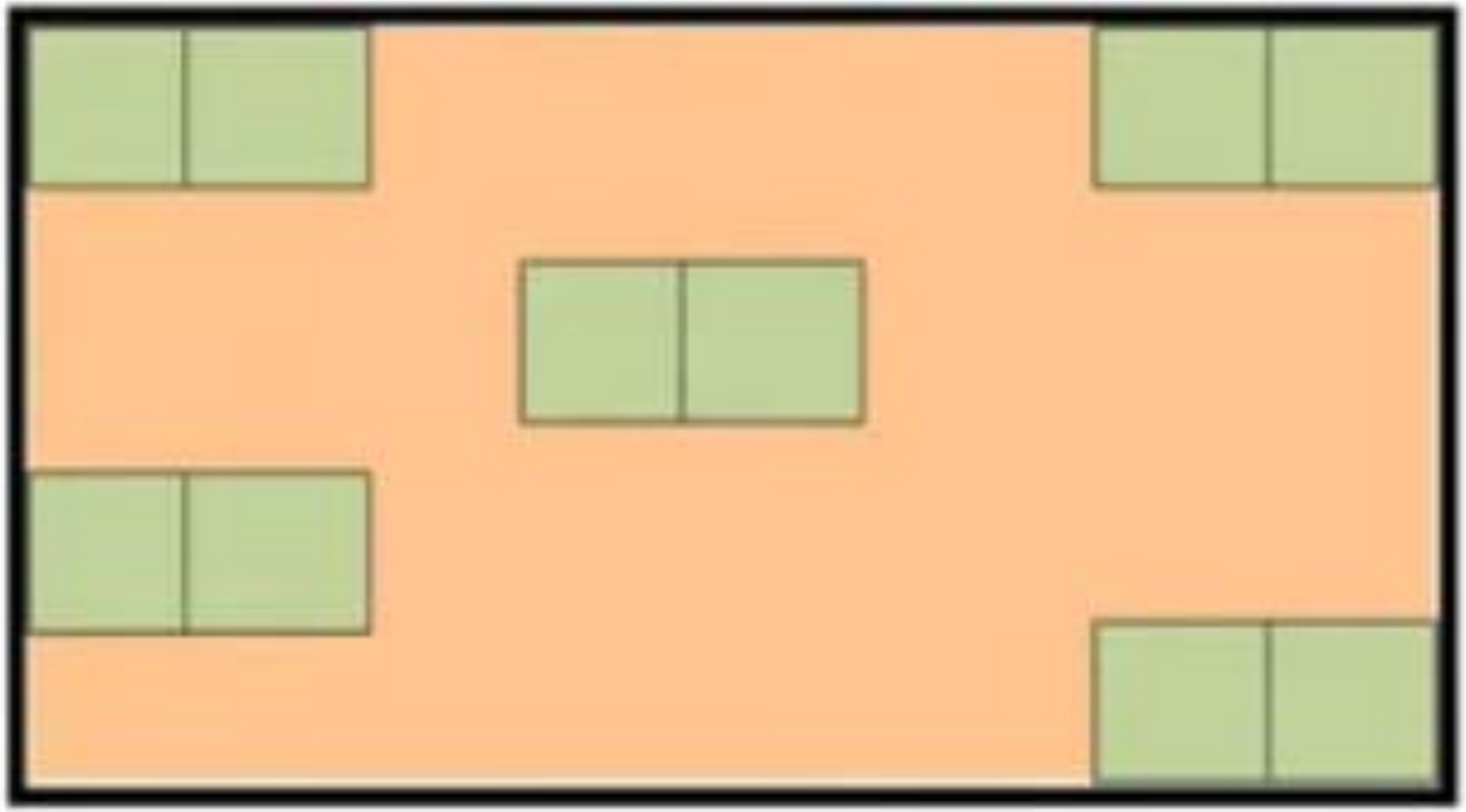
- Memory management

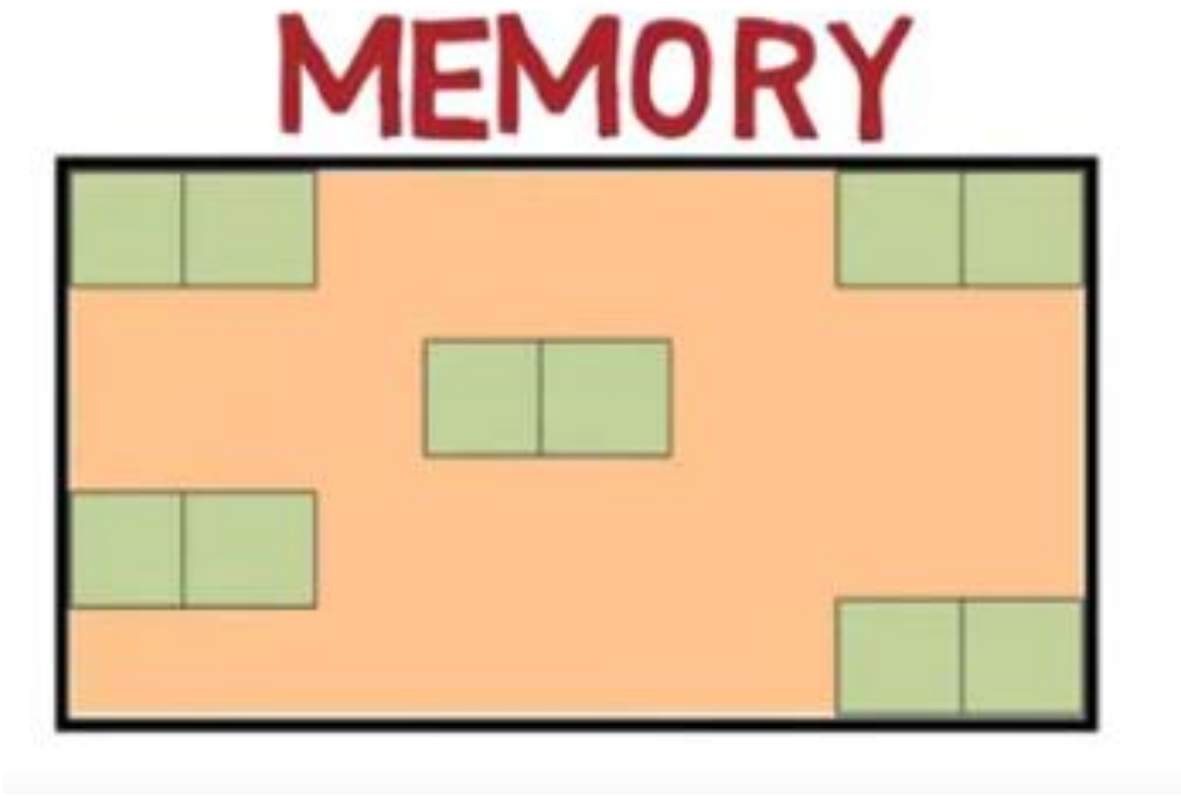
LECTURE 1

MEMORY

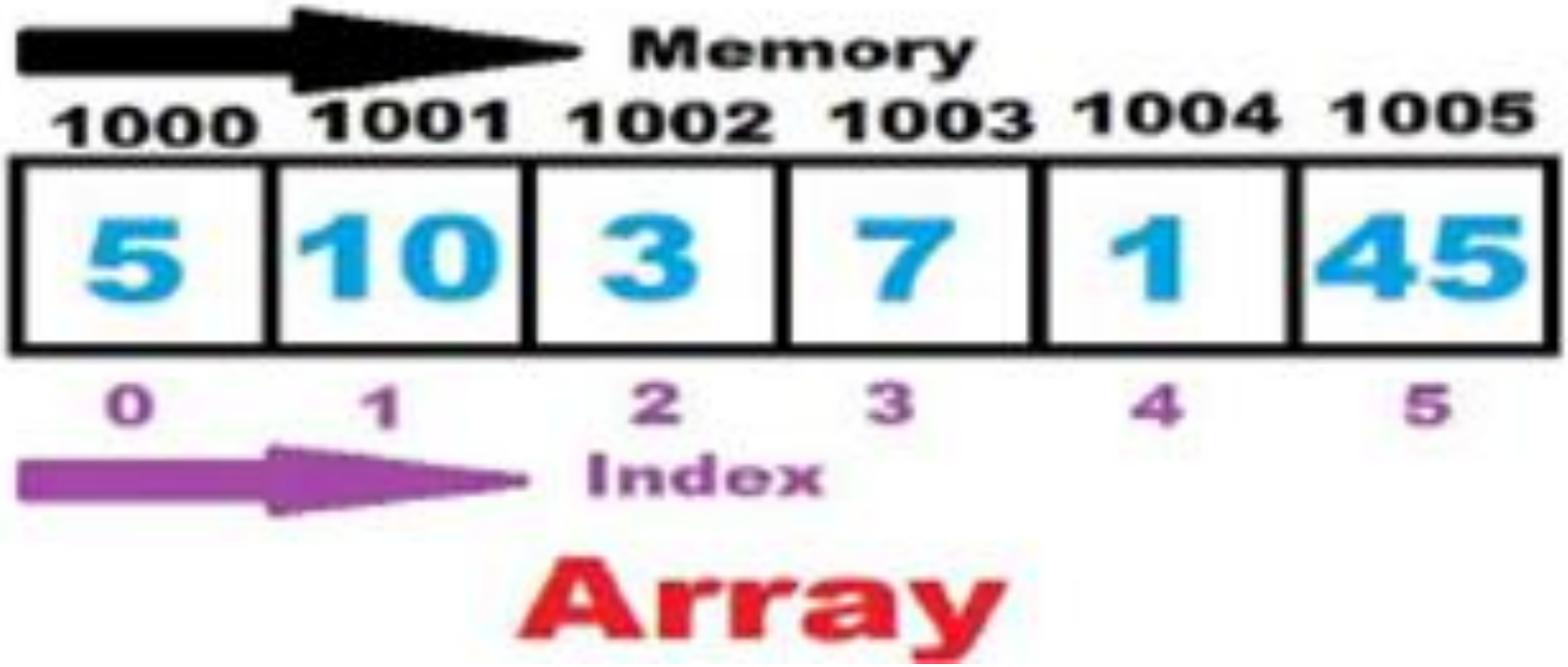


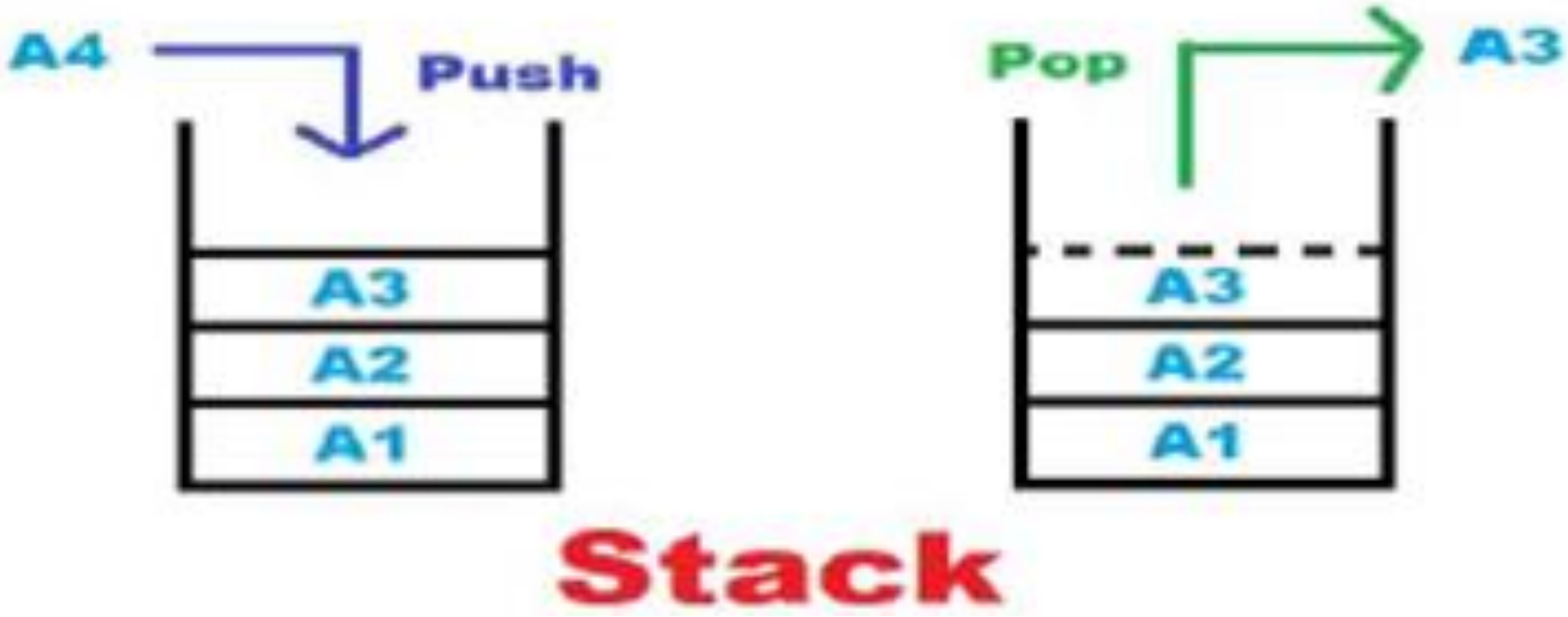
MEMORY





A data structure is the organization
of the data in a way
that it can be used efficiently


Memory
1000 1001 1002 1003 1004 1005
5 10 3 7 1 45
0 1 2 3 4 5
Index
Array


A4
Push
A3
A2
A1
Pop
A3
A3
A2
A1
Stack

The functions on the DATA

Storing

Access and Operations

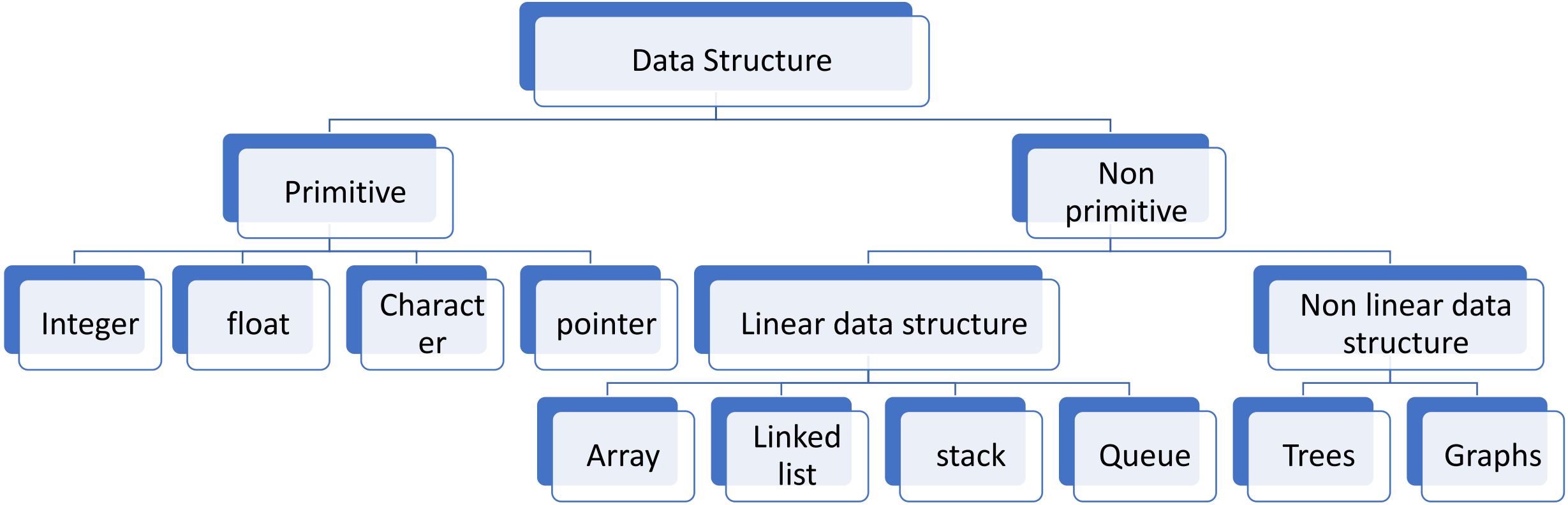
Insertion

Deletion

Searching

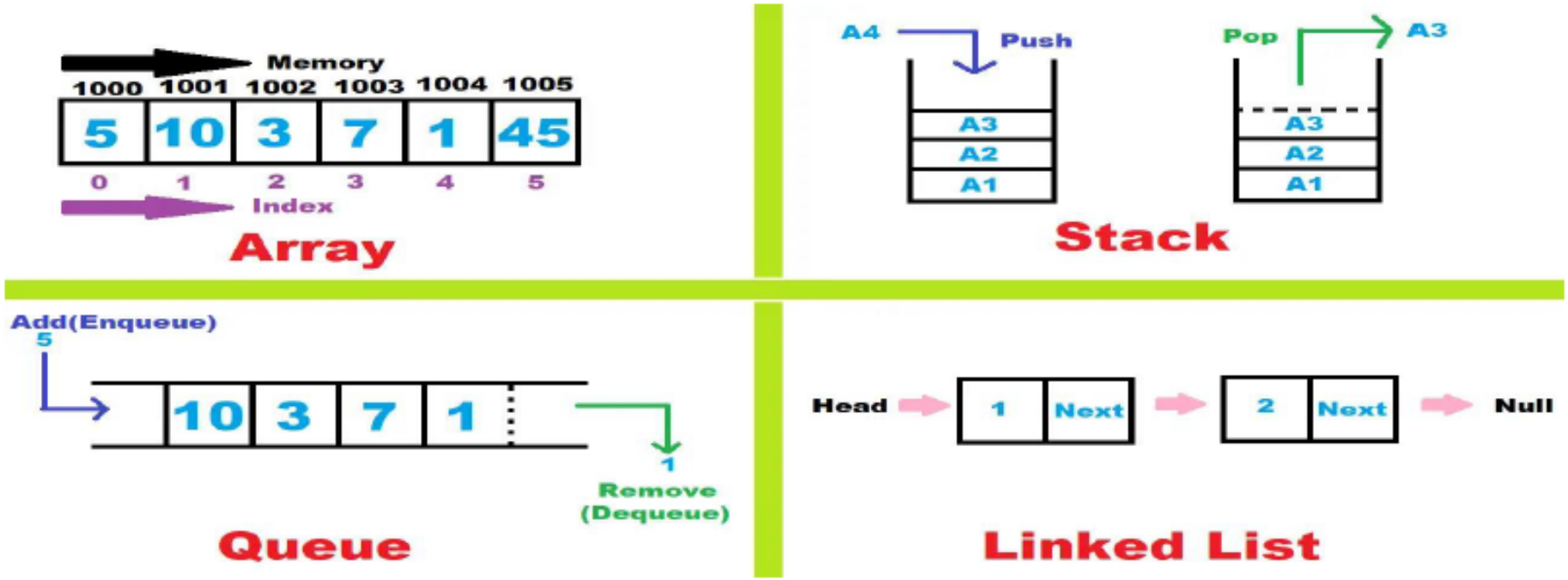
Data Structure

- A data structure is the organization of the data in away that it can be used efficiently
- Data structure affects the design of both structural & functional aspects of a program
- The program which implements the data structure is known as the implementation.
- The program that uses data structure is called a client program
- Program=algorithm + Data Structure
- Algorithm is a step by step procedure to solve a particular function.


Data Structure
Primitive
Non primitive
Integer
float
Charact er
pointer
Linear data structure
Non linear data structure
Array
Linked list
stack
Queue
Trees
Graphs

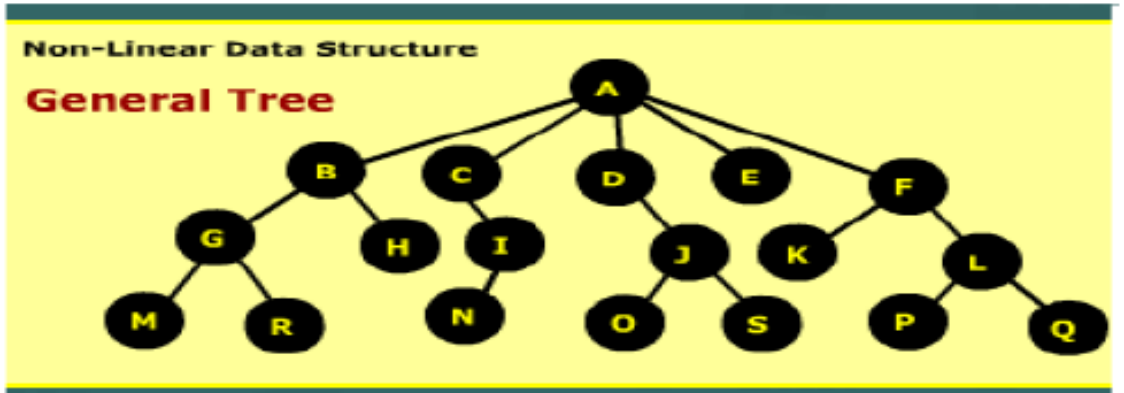
Linear Data Structure

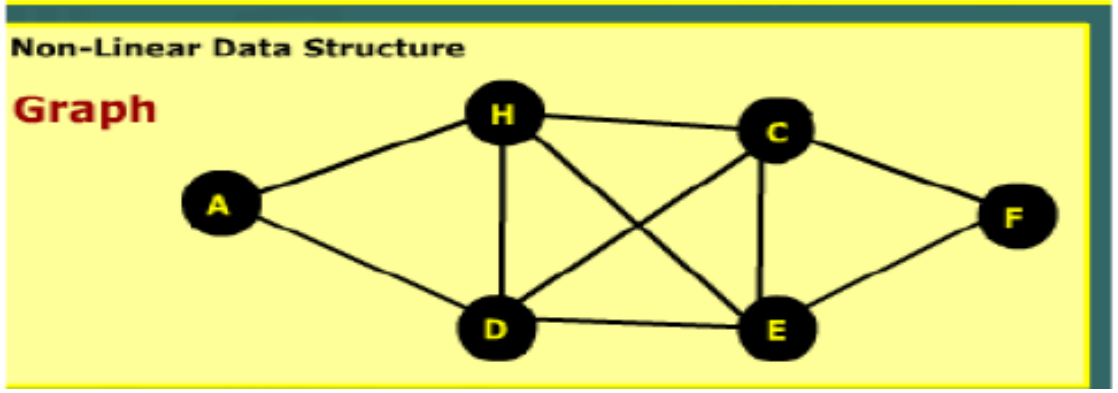
- A data structure is linear when all the elements are arranged in a linear (sequential) order.
- Each element is having only one predecessor and one successor. There is one exception the first element has no predecessor and last element has no successor.



Non-Linear Data Structure

- A data structure is non linear when all the elements are not arranged in a linear (sequential) order. There is no linear arrangement of the elements.

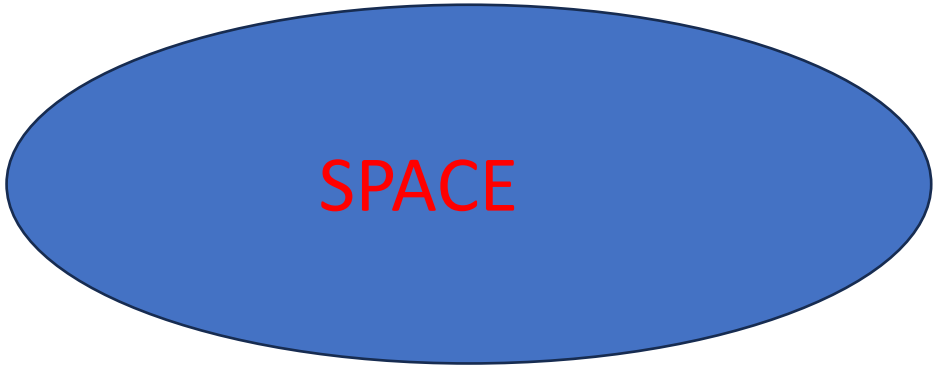


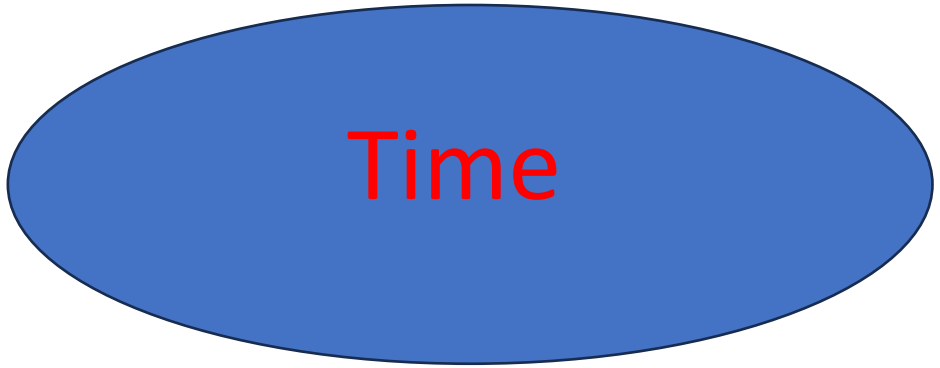


Efficiency

- Efficiency of data structures is always measured in terms of Time and Space.
- An ideal data structure could be the one that takes the least possible time for all its operations and consumes the least memory space.
- We compare the time complexity of the data structures based of operations performed on them.
- With time complexity we would be able to know which operation on which data structure will take less time.

Performance of the programming code





EXAMPLE

- Want to sum numbers form 1,…..,n
- If n= 5 → sum = 1+2+3+4+5=15

Which is the best??
Performance???

```
For (i=1; i<=n;i++)
Sum = sum+I;
```

N*(n+1)/2

Complexity

Search

10	20	16	30	5	10	21	8	4

If we want to search 10 → we find it in fast time
If we want to search 5 → we find it average time
If we want to search 4 → we find it slow time

Compare functions in the limit, that is, asymptotic analysis

10	20	16	30	5	10	21	8	4

Best case

Average case

worst case

Best case Omega notation , Ω

Average case Theta notation, Θ

Worst case Big O notation, O